

PROGRAMMING STATUS REPORT · WEEK 3

Pre-Thesis Demo

A 1–6 player online co-op psychological horror game

Unity 6.5 · Mirror · Epic Online Services

REPORTING WEEK

Week 3

HEAD PROGRAMMER

Mr. Kongkidakorn Chuayjang

SUB PROGRAMMER

Mr. Atiroj Kemwanicha

Status: the game is now playable online between two machines.

What the demo has to prove

1

Online co-op, 1–6 players

One host, up to five joiners, matched through Epic Online Services — no IP addresses, no port forwarding.

2

The 13 Rules of Horror

A dynamic rule system that changes what is safe to do during a run. The design thesis of the game.

3

Persistent progression

Equipment and currency survive between runs, stored encrypted on the player's own machine.

Out of scope for the demo by decision: multiple maps. One map ships now; the map pipeline is thesis work.

Who writes what

1

Mr. Kongkidakorn Chuayjang

Head Programmer

- Networking architecture (Mirror + EOS)
- Room creation, matchmaking, room browser
- Encrypted save / progression system
- Offline ↔ online dual-mode infrastructure
- Integration: making feature code network-safe

2

Mr. Atiroj Kemwanicha

Sub Programmer · Gameplay

- Player movement and first-person camera
- Interaction system (doors, levers, props)
- Health and damage model
- Downed / revive system
- Gameplay-facing UI behaviour

Where the two meet:

gameplay features are written single-player first, then rewritten by the head programmer to run under server authority. A scheduled step, not a repair.

This week the game went online



Rooms

A host creates a room on EOS; another machine finds and joins it.



Presence

Two machines share one world — both players spawn on both clients.



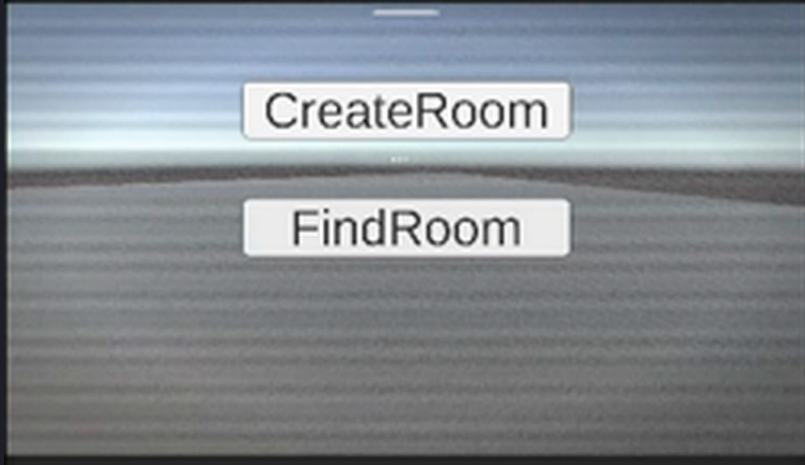
Sandbox

Solo test scenes work again with no server, so feature work is unblocked.

Where that leaves the demo

The networking risk — the part that could have failed late and taken the whole demo with it — is retired. What remains is content and feature work on a connection that has been shown to hold.

Milestone 1 — rooms on Epic Online Services



MainMenu — the two entry points under test

- Create Room → opens an EOS lobby, then starts the Mirror host
- Find Room → searches EOS and returns every open room

```
[15:35:18] เปิดห้องใหม่ ที่ Mirror เริ่มโฮสต์เกม  
UnityEngine.Debug:Log (object)  
[15:35:18] Starting server.  
UnityEngine.Debug:Log (object)  
[15:35:18] Server started listening  
UnityEngine.Debug:Log (object)
```

Editor console during the run — host side

Room config carried into the lobby:

```
map = demo_map_01 · difficulty = Normal · players = 6
```

- Room settings are published as EOS lobby attributes, so the browser can show map, difficulty and player count before joining.
- Passwords are deliberately NOT lobby attributes — EOS makes those readable by anyone searching.

Milestone 2 — two machines, one world

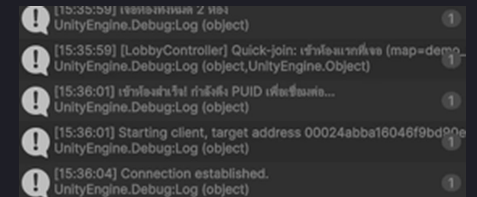


Host scene hierarchy

Two Player(Clone) objects — host and remote client — in the same scene.



Game view — the other player is visible and moving

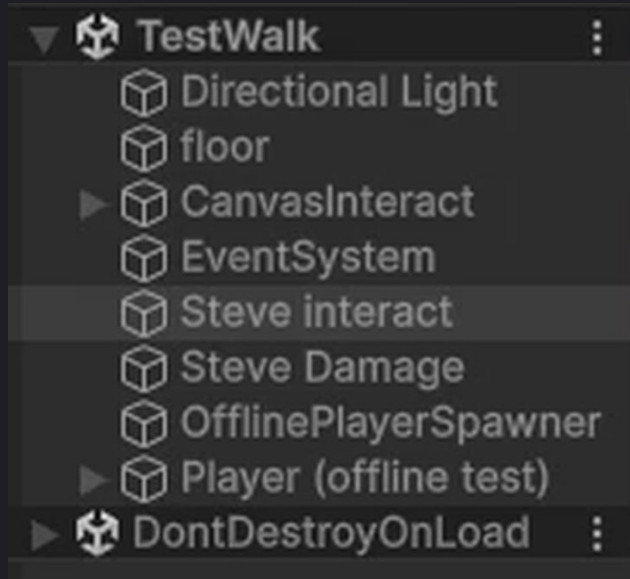


Client connection log

“Connection established” after the client resolved the host's EOS product user ID.

This is the result that was failing last week: the client connected but no player objects appeared. Root cause was a prefab registration mismatch between the two machines' checkouts.

Milestone 3 — solo testing works again



TestWalk in Play mode — nothing has vanished

The problem

Mirror force-disables every scene object carrying a `NetworkIdentity` so the server decides when to spawn it. In a sandbox scene there is no server — so interactable props disappeared the moment Play was pressed.

The fix

A dedicated offline mode. With no server and no client running, a spawner places a player and re-enables the scene objects Mirror switched off. Two earlier attempts failed on timing — Mirror's pass runs after Awake.

 [OfflinePlayerSpawner] Offline test player spawned at (2.47, 0.56, 13.29).

Editor log confirming the offline spawn

In a real match this code does nothing — it is gated on there being no server and no client.

Problems that were not obvious

SYMPTOM	WHAT WAS ACTUALLY WRONG	RESOLUTION
1 Two EOS SDKs in one build	Two different Epic integrations were installed, colliding on the same native library.	Removed the unused one. Build went to 0 errors.
2 Client saw an empty world	A server can instantiate a prefab it references; a client can only spawn one that is registered.	Identified the checkout mismatch between machines.
3 Props vanished on Play	Mirror disables scene objects after Awake, so an Awake-time fix runs too early to see it.	Rewrote the fix to watch across frames.
4 Lobby creation rejected	Epic login is asynchronous; the button could be pressed before login finished.	Gated room creation on login state.

Common thread: each was found by reading the third-party source rather than guessing at its behaviour.

Next week's plan

Mr. Kongkidakorn Chuayjang

HEAD PROGRAMMER

- Build the room browser UI on the working search
- Move the match out of the menu scene into a real level
- Private-room password exchange after join
- Start the 13 Rules system: state container and sync model

Mr. Atiroj Kemwanicha

SUB PROGRAMMER

- Extend interaction to the equipment the demo needs
- Tune downed / revive timings against playtest feel
- Author the props and triggers for the demo map

Priority call for Week 4

The 13 Rules of Horror is the design thesis and is still at zero lines. The networking foundation is finished, so it is time to stop building infrastructure and start building the game the project is actually about.

Everything built so far

Foundations

- AES-256 encrypted local save system
- Auto-load before first scene; one source of truth
- Permanent items capped at 1, lost on drop or death
- Custom Inspector for live save contents
- Editor tooling for network setup checks

Networking

- Mirror + Epic Online Services integrated and building clean
- Create room → EOS lobby → Mirror host
- Find rooms → every open room with map, difficulty, player count
- Join by Epic product user ID — no IP, no port forwarding
- Two-machine session verified: both players spawn and move
- Offline ↔ online dual mode for solo testing

Gameplay

- First-person movement and camera
- Interaction system: doors, levers, props
- Health and damage model
- Downed / revive with synced timer
- All of the above rewritten to server authority

21

C# scripts

2,821

lines of C#

5

Unity scenes

0 / 0

errors / warnings